

Noesis Lab: Project Summary

AI is making it possible to generate research ideas faster than ever, but generating more ideas does not necessarily result in faster discoveries. Researchers can still waste hours — and expensive compute — testing ideas that have already been explored, pursuing weak hypotheses, or repeating experiments without learning what to try next.

Noesis Lab helps researchers decide what is actually worth testing.

A researcher gives the system a question or hypothesis. Noesis Lab looks at existing research, identifies what is already known, highlights the gaps that remain, and recommends the most promising experiment to run next. If an experiment is worth pursuing, the system can run it, learn from the result, and use that evidence to guide the next decision.

Instead of giving researchers another AI that simply generates more ideas, Noesis Lab gives them a research process that gets smarter as it works.

We built our prototype for machine-learning researchers and AI teams, where experiments can be costly and duplicated effort adds up quickly. During the hackathon, we created a working end-to-end loop that connects published research, hypothesis evaluation, real model experiments, and the next research decision in one place.

The goal is simple: spend less time rediscovering what is already known, and more time testing what is actually new.